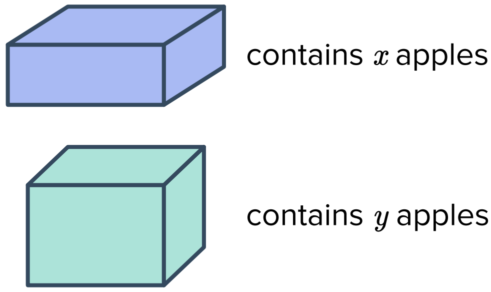
Pictures representing variables

1 The key shows how many apples are in each sized box:

Key



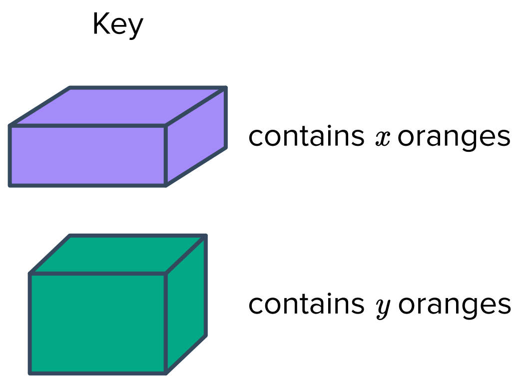
Using algebra, state how many apples are in each of the following boxes:

a

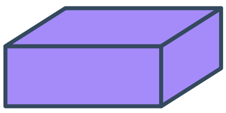
b

2 The following key shows how many oranges are in each sized box:

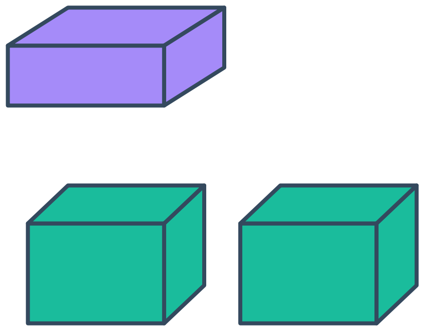
Using algebra, state how many oranges are in each of the following boxes:



a



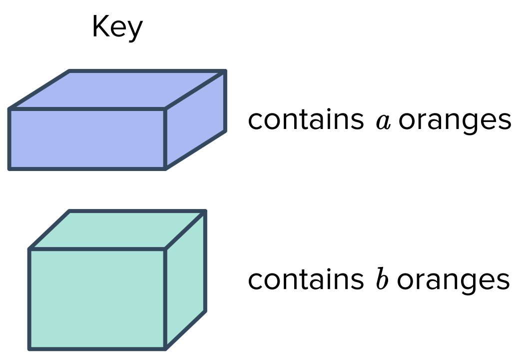
b



Adding variables

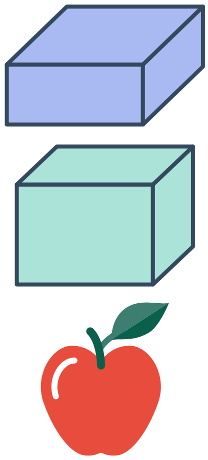
3 The given key shows how many oranges are in each size of box:

If Vincent has one box of each size, write an algebraic expression for the amount of oranges he has altogether.



4 Tom is selling some apples in boxes and some apples individually. To the right is the key showing how many apples are in each box:

Key



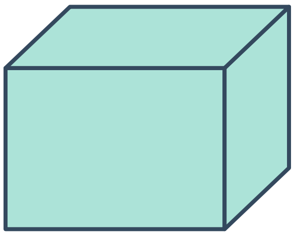
contains x apples

contains y apples

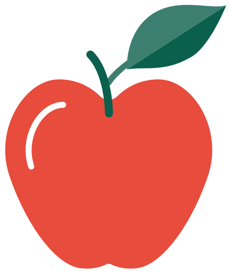
equals 1 apple

a State the number of apples that each symbol represents:

i

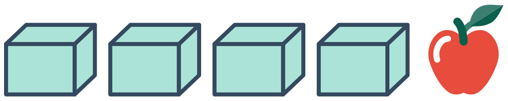


ii

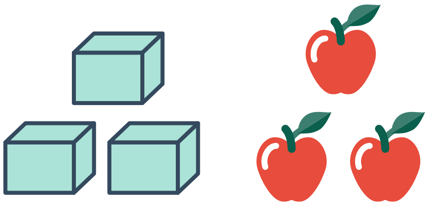


b Hence, determine the number of apples that the following images represent:

i



ii



c If there were three times as many apples in each of the images in part (b):

i Describe how the images would change.

ii What would the new algebraic expressions be for both diagrams?

Multiplying variables

5 For each of the following situations, complete the pattern in the table:

a A single box contains x apples.

Number of boxes	1	2	3	4	5
Number of apples	x	$2x$			

b A single box contains $2y+1$ oranges.

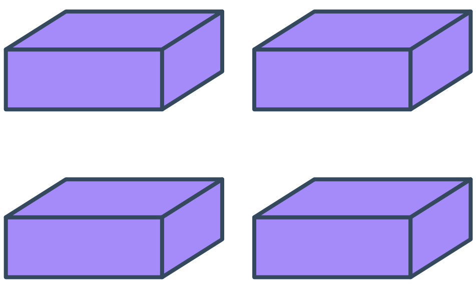
Number of boxes	1	2	3	5	10
Number of oranges	$2y+1$	$4y+2$			

c A single box contains $z+2$ bananas.

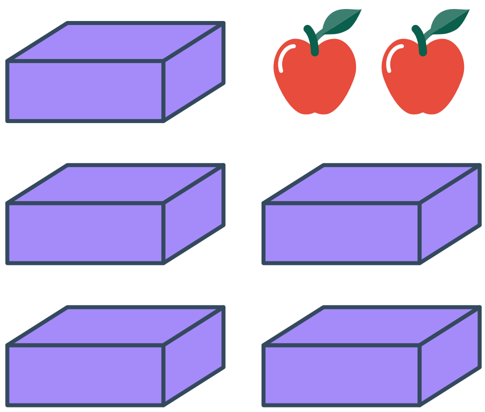
Number of boxes	1	2	3	5	10
Number of bananas	$z+2$	$2z+4$			

6 Write an algebraic expression for the number of pieces of fruit in each of the following situations:

a A single box contains x apples.

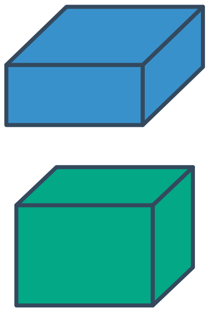


b A single box contains q apples.



7 Using the given key, state how many apples are represented in the following diagrams using algebra:

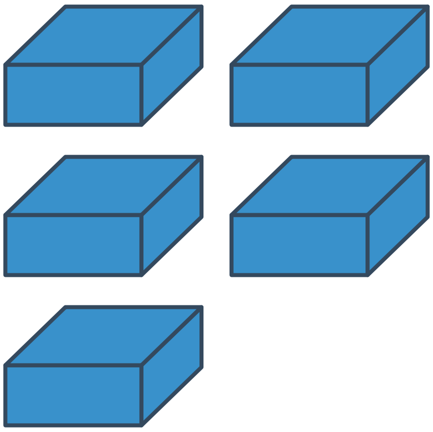
Key



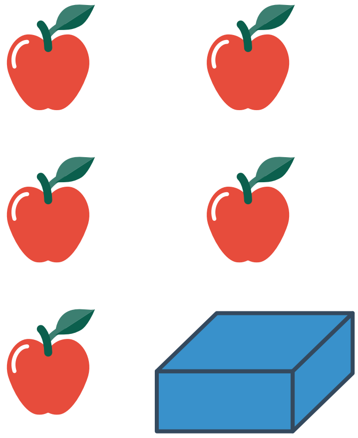
contains a apples

contains b apples

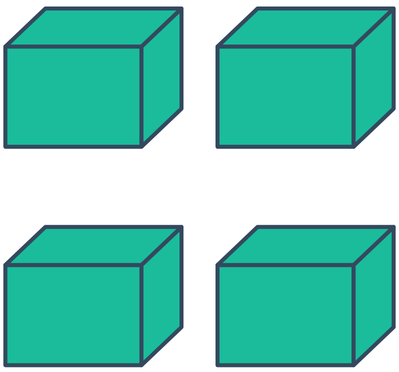
a



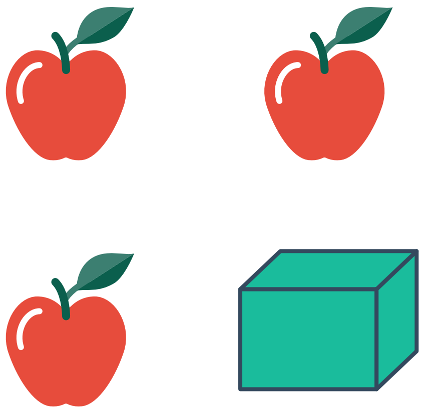
b



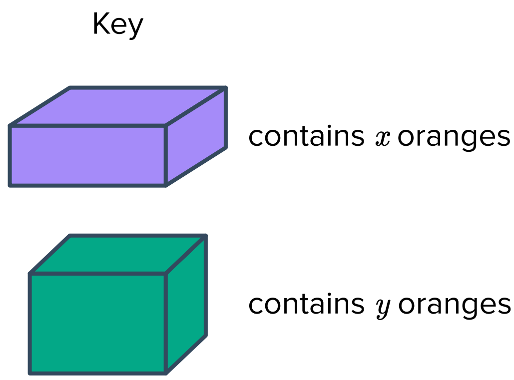
c



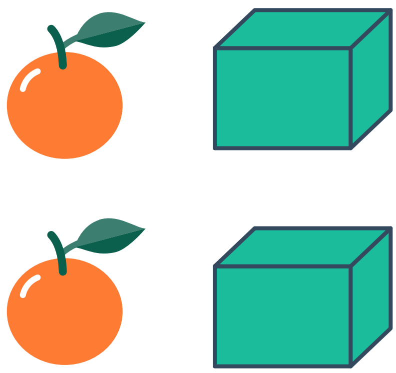
d



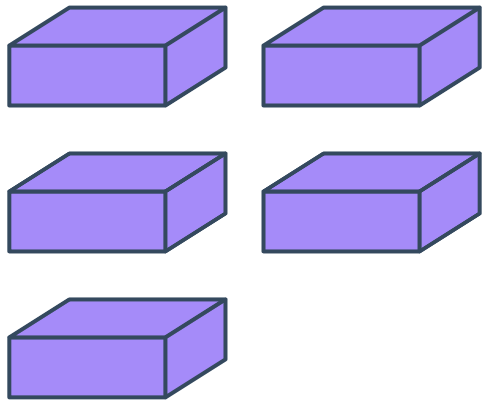
8 Using the given key, state how many oranges are represented in the following diagrams using algebra:



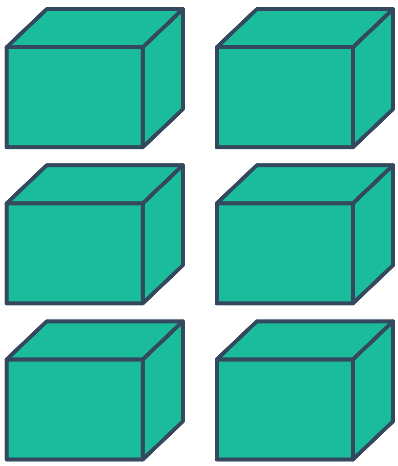
a



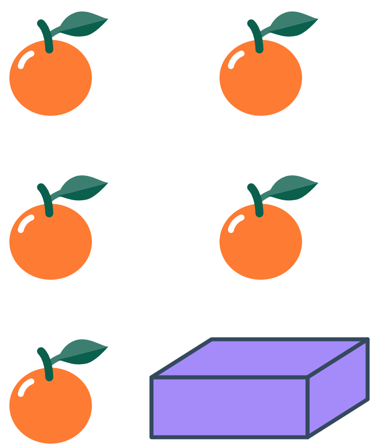
b



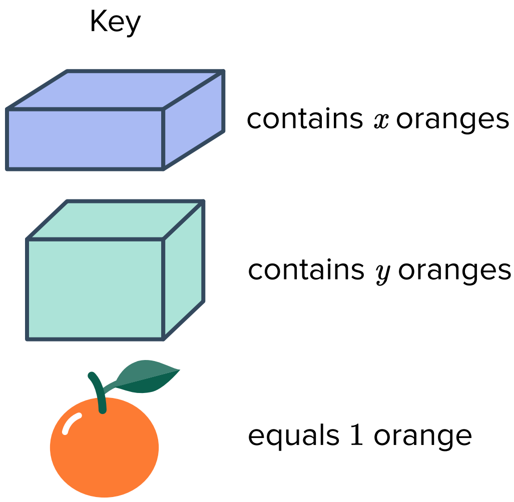
c



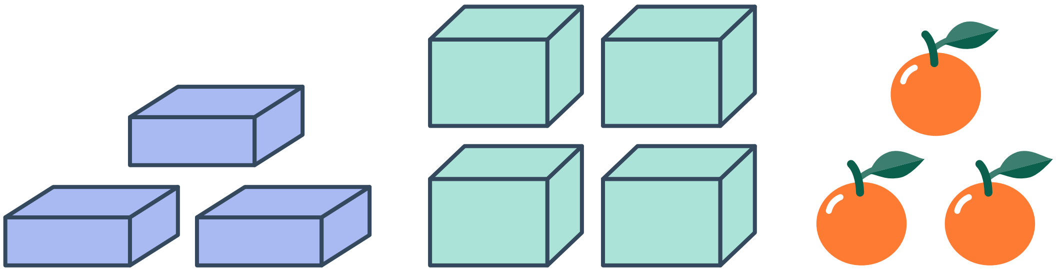
d



9 Xanthe is selling some oranges in boxes and some oranges individually. The key is provided to show how many oranges each box represents:



Use the key to state the number of oranges that the following image represents:

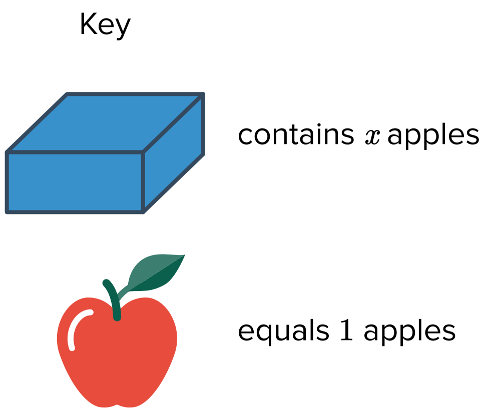


10 For each of the following situations:

i Use the given key to draw a diagram to represent the number of pieces of fruit each person has.

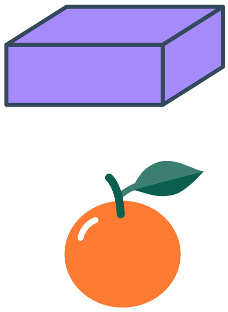
ii Write an algebraic expression for the number of apples that they have altogether.

a Georgia has three boxes containing x apples each, and is then given four more apples.



b Jimmy has two boxes of oranges containing x oranges each and is then given two more oranges.

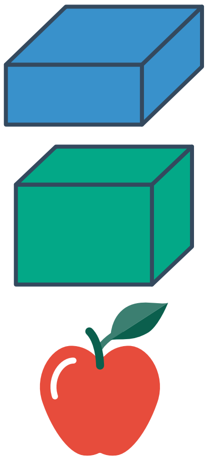
Key



11 For each situation, use the given key to draw a diagram that represents the expression:

a $2y + 3$ apples

Key



b $2a + 2$ oranges

Key